Hugo Marquerie 11/05/2025

Teorema de equivalencia entre analiticidad y holomorfía

Teorema 1 (Equivalencia entre analiticidad y holomorfía).
Sea $f\colon \Omega \subset \mathbb{C} \longrightarrow \mathbb{C}$ con Ω abierto, entonces

$$f \text{ es analítica en } \Omega \iff f \text{ es holomorfa en } \Omega.$$

***Demostración*:**

$\boxed{\Rightarrow}$

$\boxed{\Leftarrow}$ Sea $f \in \mathcal{H}(\Omega)$ y $z_0 \in \Omega$. Consideramos $z > 0$ tal que $\overline{D(z_0, r)} \subset \Omega$. Si denotamos $\partial D(z_0, r)^+ = C_r^+$ entonces, por la fórmula integral de Cauchy, $\forall z \in D(z_0, r)$:

$$\begin{aligned} f(z) &= \frac{1}{2\pi i}\int_{C_r^+} \frac{f(w)}{w-z}\,\mathrm{d}w = \frac{1}{2\pi i}\int_{C_r^+} \frac{f(w)}{(w-z_0)-(z-z_0)}\,\mathrm{d}w \\ &= \frac{1}{2\pi i}\int_{C_r^+} \frac{f(w)}{(w-z_0)\left(1-\frac{z-z_0}{w-z_0}\right)}\,\mathrm{d}w = \frac{1}{2\pi i}\int_{C_r^+} \frac{f(w)}{(w-z_0)}\cdot\frac{1}{\left(1-\frac{z-z_0}{w-z_0}\right)}\,\mathrm{d}w. \end{aligned}$$

Observamos que $\left(1-\dfrac{z-z_0}{w-z_0}\right)^{-1} = \displaystyle\sum_{n=0}^{\infty}\left(\frac{z-z_0}{w-z_0}\right)^n$ para $\left|\dfrac{z-z_0}{w-z_0}\right| < 1$.

Pero como $w \in C_r^+$, tenemos que $\left|\frac{z-z_0}{w-z_0}\right| = \frac{z-z_0}{r} < 1$ y, por tanto, la serie converge uniformemente en C_r^+. Entonces, podemos permutar la integral y el sumatorio:

$$\begin{aligned} \forall z \in D(z_0, r) : f(z) &= \frac{1}{2\pi i}\int_{C_r^+} \frac{f(w)}{(w-z_0)}\cdot\left[\sum_{n=0}^{\infty}\left(\frac{z-z_0}{w-z_0}\right)^n\right]\mathrm{d}w \\ &= \sum_{n=0}^{\infty}\underbrace{\frac{1}{2\pi i}\int_{C_r^+}\frac{f(w)}{(w-z_0)^{n+1}}\,\mathrm{d}w}_{a_n}\cdot(z-z_0)^n = \sum_{n=0}^{\infty} a_n\,(z-z_0)^n. \end{aligned}$$

■

Referenciado en

- `teo-formula-integral-cauchy-derivadas`

Temporary page!

LaTeX was unable to guess the total number of pages correctly. As there was some unprocessed data that should have been added to the final page this extra page has been added to receive it.

If you rerun the document (without altering it) this surplus page will go away, because LaTeX now knows how many pages to expect for this document.